

Notes: Arellano & Bond (RES, 1991)

Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Generated data for Monte Carlo experiments	2
2.2	Real data: U.K. company employment panel	2
2.3	Sample selection and variables	2
3	Models and empirical specifications	3
3.1	Baseline dynamic panel model	3
3.2	Moment restrictions in first differences	3
3.3	GMM estimator	3
3.4	Specification tests	3
3.5	Employment equation	3
4	Identification and empirical strategy	4
4.1	Core identification strategy	4
4.2	Why AR(2) in differences is central	4
4.3	Employment application identification	4
5	Tables and figures	4
5.1	Table 1: Biases in the estimates	4
5.2	Table 2: Sizes of the test statistics — number of rejections out of 100 cases	4
5.3	Table 3: Power of the test statistics and heteroskedasticity robustness	5
5.4	Table 4: Employment equations — GMM estimates	5
5.5	Table 5: Employment equations — alternative estimates	6
5.6	Table A1: Number of records on each company	6
6	Interpretive synthesis	6
7	Short summary	7
8	Conclusion	7
8.1	Core conclusion	7
8.2	Practical conclusion	8
8.3	Takeaway	8

1 Big picture

- **Question.** How should researchers estimate and test dynamic panel data models with fixed effects, lagged dependent variables, predetermined regressors, and no strictly exogenous instruments?
- **Core contribution.** The paper develops the difference-GMM estimator now known as the Arellano–Bond estimator and gives practical specification tests.
- **Key identifying idea.** First-differencing removes individual effects, and lagged levels dated two periods or more become valid instruments if level errors are not serially correlated.
- **Main tests.** The paper studies a direct test of second-order serial correlation, Sargan tests, difference-Sargan tests, and Hausman tests.
- **Monte Carlo message.** GMM has small bias and much lower variance than Anderson–Hsiao IV; robust serial-correlation tests and two-step difference-Sargan tests perform well; two-step Hausman tests over-reject.
- **Empirical application.** The paper estimates dynamic employment equations using an unbalanced panel of 140 U.K. manufacturing firms from 1976–1984.

2 Data and measurement

2.1 Generated data for Monte Carlo experiments

The simulation uses $N = 100$, $T = 7$, and 100 replications. The dependent variable is generated from a dynamic panel model with an individual effect, a lagged dependent variable, and an exogenous regressor. Alternative designs introduce MA(1) serial correlation and heteroskedasticity.

Interpretation. The simulations evaluate both estimator bias and the finite-sample size and power of the specification tests.

2.2 Real data: U.K. company employment panel

The empirical application uses published accounts of 140 quoted U.K. manufacturing companies during 1976–1984. The panel is unbalanced: some firms have seven observations, others eight or nine. The GMM estimates use 611 observations over 1979–1984 after differencing and constructing lags.

Interpretation. This is the canonical large- N , small- T firm panel setting for difference-GMM.

2.3 Sample selection and variables

The sample requires complete accounting records and excludes firms with large accounting-year changes or implausible jumps in employment, real wages, real capital, real inventories, or real sales. Employment is the log of U.K. employees; real wages are remuneration per employee adjusted for hours and deflated by industry price deflators; capital is replacement-cost gross fixed assets; industry output is value added in nine manufacturing sub-sectors.

Interpretation. The variables map the dynamic panel estimator to a labour-demand equation with firm effects, adjustment dynamics, wages, capital, and demand shifters.

3 Models and empirical specifications

3.1 Baseline dynamic panel model

$$y_{it} = \alpha y_{i,t-1} + \eta_i + v_{it}, \quad |\alpha| < 1.$$

Interpretation. OLS in levels is upward biased because $y_{i,t-1}$ is correlated with η_i ; within-groups is biased when T is small.

3.2 Moment restrictions in first differences

$$\begin{aligned} \Delta y_{it} &= \alpha \Delta y_{i,t-1} + \Delta v_{it}. \\ E[(\Delta y_{it} - \alpha \Delta y_{i,t-1}) y_{i,t-j}] &= 0, \quad j = 2, \dots, t-1; \quad t = 3, \dots, T. \end{aligned}$$

Interpretation. Lagged levels dated two periods or more are valid instruments because they are orthogonal to the differenced error under no serial correlation in level errors.

3.3 GMM estimator

$$\hat{\beta} = \arg \min_{\beta} v(\beta)' Z A_N Z' v(\beta).$$

One-step GMM uses a simple homoskedastic weight matrix. Two-step GMM updates the weighting matrix using residuals and is asymptotically efficient.

Interpretation. Two-step standard errors can be downward biased in finite samples, so coefficient precision must be interpreted cautiously.

3.4 Specification tests

- **m_2 test:** test for second-order serial correlation in differenced residuals.
- **Sargan test:** test of all overidentifying restrictions.
- **Difference-Sargan test:** test of extra instruments valid only under stronger assumptions.
- **Hausman test:** comparison of estimators based on different instrument sets.

Interpretation. The m_2 test is the most direct diagnostic for the validity of lagged-level instruments.

3.5 Employment equation

$$n_{it} = \alpha_1 n_{i,t-1} + \alpha_2 n_{i,t-2} + \beta(L) x_{it} + \lambda_t + \eta_i + v_{it}.$$

Here x_{it} includes real wages, gross capital, and industry output.

Interpretation. The equation can be read as a dynamic labour-demand model or, under stronger restrictions, as an Euler equation.

4 Identification and empirical strategy

4.1 Core identification strategy

First-differencing removes firm fixed effects. Lagged levels instrument differenced endogenous variables. The identifying assumption is that level errors are not serially correlated.

Interpretation. Identification is internal: it comes from lag timing and orthogonality restrictions, not outside instruments.

4.2 Why AR(2) in differences is central

First-order serial correlation in differenced residuals is expected even when level errors are white noise. Second-order serial correlation in differenced residuals indicates serial correlation in level errors and invalidates lagged instruments.

Interpretation. This is why the Arellano–Bond AR(2) test became standard in dynamic panel work.

4.3 Employment application identification

Lagged employment levels instrument differenced lagged employment. Wages and capital are first treated as strictly exogenous, then alternatively as endogenous using deeper lags. Time dummies absorb aggregate shocks; industry output controls for sectoral demand.

Interpretation. The empirical application shows how changing timing assumptions changes the instrument set and the interpretation of diagnostic tests.

5 Tables and figures

The paper has no graphical figures. The key evidence is in tables, directly cropped from the original paper.

5.1 Table 1: Biases in the estimates

Read: GMM estimates are close to the truth: for true $\alpha = 0.5$, GMM1 and GMM2 estimate 0.4884 and 0.4920, while OLS gives 0.7216 and within-groups gives 0.3954. Anderson–Hsiao can be very imprecise or nearly unidentified.

Interpretation. The table shows why difference-GMM improves on both least-squares and simple IV estimators.

Economic message. The estimator is designed for the common case where OLS and within estimates are biased in opposite directions.

5.2 Table 2: Sizes of the test statistics — number of rejections out of 100 cases

Read: Under the null, the m_2 , Sargan, and difference-Sargan tests are reasonably approximated by their asymptotic distributions; the two-step Hausman statistic over-rejects.

over time and homoskedastic: $\theta_1 = \phi = 0$, $\theta_0 = \sigma^2 = 1$, $\sigma_v^2 = 1$, $\beta = 1$, $\rho = 0.8$ and $\sigma_v^2 = 0.9$. Tables 1 and 2 summarize the results for $\alpha = 0.2, 0.5, 0.8$ obtained from 100 replications. Results for other variants of this design were calculated ($N = 200$, $T = 6$, $\sigma^2 = 2$, 5 , $\sigma_v^2 = 0$, $\rho = 0$), and are available from the authors on request. However the conclusions are the same as for the results reported here.

Table 1 reports sample means and standard deviations for one-step and two-step GMM estimators (GMM1 and GMM2 respectively), OLS in levels, within-groups, and

TABLE 1
Biases in the estimates

	GMM1	GMM2	OLS	Within-groups	AHd	AHI	One-step ASE	Robust One-step ASE	Two-step ASE
$\alpha = 0.5, \beta = 1$									
Coefficient: α									
Mean	0.4884	0.4920	0.7216	0.3954	-2.4753	0.5075	0.0683	0.0677	0.0604
St. Dev.	0.0671	0.0739	0.0216	0.0272	45.9859	0.0821	0.0096	0.0120	0.0106
Coefficient: β									
Mean	1.0053	0.9976	0.7002	1.0409	0.1625	0.9996	0.0612	0.0607	0.0548
St. Dev.	0.0631	0.0668	0.0484	0.0480	9.8406	0.0650	0.0031	0.0055	0.0052
$\alpha = 0.2, \beta = 1$									
Coefficient: α									
Mean	0.1937	0.1979	0.5108	0.0957	0.2025	0.2044	0.0610	0.0602	0.0533
St. Dev.	0.0597	0.0670	0.0340	0.0309	0.1973	0.0661	0.0045	0.0066	0.0060
Coefficient: β									
Mean	1.0048	0.9960	0.7030	1.0430	0.9973	0.9991	0.0620	0.0615	0.0553
St. Dev.	0.0630	0.0687	0.0526	0.0476	0.0818	0.0654	0.0028	0.0058	0.0052
$\alpha = 0.8, \beta = 1$									
Coefficient: α									
Mean	0.7827	0.7810	0.8997	0.7160	0.8103	0.8038	0.0529	0.0527	0.0470
St. Dev.	0.0582	0.0609	0.0090	0.0206	0.1313	0.2677	0.0069	0.0082	0.0075
Coefficient: β									
Mean	1.0001	0.9926	0.7754	1.0137	1.0000	0.9980	0.0609	0.0601	0.0544
St. Dev.	0.0622	0.0671	0.0423	0.0461	0.0789	0.0893	0.0035	0.0056	0.0056

Notes.
(i) $N = 100$, $T = 7$, 100 replications, $\sigma^2 = \sigma_v^2 = 1$.
(ii) Exogenous variable is first order autoregressive with $\rho = 0.8$ and $\sigma_v^2 = 0.9$.
(iii) GMM1 and GMM2 are respectively one step and two step difference—IV estimators of the type described in Section 2. Both GMM use $Z_s = [\text{diag}(y_{11} \dots y_{1T}) : (\bar{x}_{11} \dots \bar{x}_{1T})']$ ($s = 1, \dots, T-2$).
(iv) AHd and AHI are the Anderson-Hsiao stacked—IV estimators of the equation in first differences that use $\Delta y_{1(t-2)}$ and $y_{1(t-2)}$ as an instrument for $\Delta y_{1(t-1)}$ respectively.
(v) One Step ASE and Robust One Step ASE are estimates of the asymptotic standard errors of GMM1. The former are only valid for i.i.d. errors while the latter are robust to general heteroskedasticity over individuals and over time. Two step ASE is a robust estimate of the asymptotic standard errors of GMM2.

6. The optimal instrument set for the system of first difference equations would be $z_t = \text{diag}(y_{11} \dots y_{1T}, x_{11}, x_{12}, \dots, x_{1T})$.

TABLE 2

Sizes of the Test Statistics, Number of rejections out of 100 cases

	One-step m_2	Robust one-step m_2	Two-step m_2	One-step Sargan (df=14)	Two-step Sargan (df=14)	One-step difference-Sargan (df=5)	Two-step difference-Sargan (df=5)	One-step Hausman (df=1)	Two-step Hausman (df=1)
$\alpha = 0.5$									
10	12	15	14	11	7	13	12	13	20
5	5	5	6	6	4	6	6	9	12
1	1	1	1	0	0	2	2	2	3
Mean	0.013	0.003	0.002	13.622	13.644	5.195	5.209	1.135	1.487
Variance	1.049	1.071	1.063	28.004	27.092	10.889	10.788	3.547	4.493
$\alpha = 0.2$									
10	12	13	13	8	6	12	9	11	17
5	4	5	4	4	3	8	6	7	13
1	2	1	1	0	0	1	1	2	3
Mean	-0.002	-0.010	-0.018	13.519	13.601	5.043	5.004	1.047	1.528
Variance	1.127	1.146	1.124	24.471	22.002	10.928	10.096	2.612	5.003
$\alpha = 0.8$									
10	12	15	15	9	10	7	12	11	12
5	5	6	6	7	4	4	9	5	8
1	1	0	0	1	2	1	2	0	4
Mean	0.047	0.037	0.036	13.423	13.883	4.917	5.199	0.928	1.239
Variance	1.004	1.029	1.028	31.492	28.854	8.916	11.333	1.648	4.325

Notes.
(i) $N = 100$, $T = 7$, 100 replications, $\sigma^2 = \sigma_v^2 = 1$, $\beta = 1$.
(ii) Exogenous variable is first order autoregressive with $\rho = 0.8$ and $\sigma_v^2 = 0.9$.
(iii) All tests are based on the one- and two-step GMM estimation reported in Table 1 as well as on restricted versions of those which only use the columns of Z_s that remain valid instruments when the errors are MA (1). The one-step m_2 statistic is described in the Appendix.
(iv) Results on an extended set of experiments are available from the authors on request.

Interpretation. The table calibrates which specification tests are reliable in finite samples.

Economic message. Researchers should report tests, but they should also know which tests have poor finite-sample behavior.

5.3 Table 3: Power of the test statistics and heteroskedasticity robustness

Read: The m_2 test has high power against MA(1) serial correlation and robust versions behave better under heteroskedasticity. Non-robust one-step tests can reject too often.

Interpretation. The table shows that serial-correlation and overidentification diagnostics are complementary.

Economic message. Robust testing is essential when firm data are heteroskedastic.

5.4 Table 4: Employment equations — GMM estimates

Read: Lagged employment is strongly persistent, wage effects are negative, capital and industry output effects are mostly positive, and robust m_2 statistics do not reject.

Interpretation. The table illustrates the full Arellano–Bond workflow: estimate in differences, use internal instruments, and report diagnostic tests.

Economic message. The labour-demand estimates are plausible, but their credibility rests on the specification tests.

TABLE 3
Power of the Test Statistics, Number of rejections out of 100 cases

	One-step m_2	Robust one-step m_2	Two-step m_2	One-step Sargan (df = 14)	Two-step Sargan (df = 14)	One-step difference- Sargan (df = 5)	Two-step difference- Sargan (df = 5)	One-step Hausman (df = 1)	Two-step Hausman (df = 1)
<i>Serial correlation: Corr (ϵ_{it-1}) = 0.2</i>									
10	54	53	53	47	33	60	46	27	25
5	45	46	46	33	20	53	34	15	20
1	24	25	25	22	2	29	9	8	11
Mean	-1.833	-1.823	-1.823	20.673	17.988	11.307	9.151	2.076	2.431
Variance	1.079	0.987	0.988	70.881	34.447	40.017	20.773	6.855	12.496
<i>Serial correlation: Corr (ϵ_{it-1}) = 0.3</i>									
10	95	96	96	78	71	91	86	29	33
5	91	92	92	72	47	86	70	20	22
1	78	77	78	57	30	69	51	11	11
Mean	-3.293	-3.118	-3.121	32.037	24.180	22.112	15.314	2.537	2.707
Variance	1.033	0.995	0.705	135.543	40.127	94.261	30.339	12.430	13.640
<i>Heteroskedasticity: Var (ϵ_{it}) = x_{it}^2, ϵ_{it} AR (1) data</i>									
10	23	9	10	43	8	35	7	24	25
5	15	2	2	35	0	24	4	16	16
1	3	0	0	15	0	15	0	8	5
Mean	0.029	-0.023	-0.038	20.967	14.495	9.304	2.216	2.158	1.938
Variance	1.708	1.005	0.991	70.077	17.194	36.410	8.737	13.954	6.355
<i>Heteroskedasticity: Var (ϵ_{it}) = x_{it}^2, ϵ_{it} U.K. Sales</i>									
10	71	6	7	100	0	93	13	74	81
5	68	3	2	100	0	92	3	69	76
1	54	0	0	100	0	88	1	62	72
Mean	-0.870	-0.225	-0.301	207.789	14.940	63.389	5.611	40.241	154.452
Variance	15.022	1.126	1.014	5863.623	5.405	2659.712	9.095	12059.642	339656.703

Notes:
(i) $N = 100$, $T = 7$, 100 replications, $\alpha = 0.5$, $\beta = 1$, $\sigma^2 = 0$.
(ii) In the serial correlation designs ϵ_{it} has been generated as MA (1) with standard normal random errors.
(iii) In the first three designs x_{it} is AR (1) with $\rho = 0.8$ and $\sigma^2 = 0.9$.
(iv) In the fourth design, x_{it} are total sales from a sample of quoted U.K. companies scaled to have $\text{Var}(\Delta x_{it}) = 1$. x_{it} behaves as a random walk process with large differences in mean and variance across firms.
(v) In the heteroskedastic designs there is no serial correlation in the ϵ_{it} .

5.5 Table 5: Employment equations — alternative estimates

Read: Anderson–Hsiao estimates are very imprecise; OLS in levels has upward bias in the lagged dependent variable; within-groups is closer but still subject to dynamic panel bias.

Interpretation. Table 5 explains why using the full GMM moment set is valuable.

Economic message. Arellano–Bond is a practical middle ground between biased least squares and inefficient simple IV.

5.6 Table A1: Number of records on each company

Read: The panel contains 103 firms with seven observations, 23 firms with eight observations, and 14 firms with nine observations.

Interpretation. The unbalanced-panel extension allows the application to use more data than a balanced panel would.

Economic message. Dynamic panel methods must work with realistic, uneven firm histories.

6 Interpretive synthesis

The paper’s contribution is an estimator-plus-testing protocol. Difference-GMM uses internal instruments from lagged levels. The m_2 test verifies the core serial-correlation condition. Sargan and difference-Sargan

Employment equations
GMM estimates (all variables in first differences)

Dependent variable: $\ln(\text{Employment})_{it}$ Sample Period: 1979–1984 (140 companies)

Independent variables	(a1)	(a2)	(b)	Instrumenting wages and capital* (c)	(d)
m_{it-1}	0.686 (0.145)	0.629 (0.090)	0.474 (0.085)	0.800 (0.048)	0.825 (0.056)
m_{it-2}	-0.085 (0.056)	-0.065 (0.027)	-0.053 (0.027)	-0.116 (0.021)	-0.074 (0.020)
w_{it}	-0.608 (0.178)	-0.526 (0.054)	-0.513 (0.049)	-0.640 (0.054)	—
w_{it-1}	0.393 (0.168)	0.311 (0.094)	0.225 (0.080)	0.564 (0.066)	0.431 (0.076)
k_{it}	0.357 (0.059)	0.278 (0.045)	0.293 (0.039)	0.220 (0.051)	—
k_{it-1}	-0.058 (0.073)	0.014 (0.053)	—	—	-0.077 (0.045)
k_{it-2}	-0.020 (0.033)	-0.040 (0.026)	—	—	—
y_{80}	0.608 (0.172)	0.592 (0.116)	0.610 (0.109)	0.890 (0.098)	—
y_{81-1}	-0.711 (0.232)	-0.566 (0.140)	-0.446 (0.125)	-0.875 (0.105)	-0.115 (0.100)
y_{81-2}	0.106 (0.141)	0.101 (0.113)	—	—	0.096 (0.092)
m_2	-0.516	-0.434	-0.327	-0.610	-1.259
Sargan test	65.8 (25)	31.4 (25)	30.1 (25)	63.0 (50)	68.3 (51)
Difference-Sargan	41.9 (6)	15.4 (6)	10.0 (6)	28.6 (20)	31.6 (20)
Hausman	5.8 (1)	14.4 (1)	13.4 (1)	2.0 (1)	2.9 (1)
Wald test	408.3 (10)	667.0 (10)	372.0 (7)	779.3 (7)	623.9 (6)
No. of observations	611	611	611	611	611

* A subset of valid moment restrictions involving lagged wages and capital are exploited—see note (vi). Additional instruments used are the stacked levels and first differences of (firm real sales) $_{(t-2)}$ and (firm real stocks) $_{(t-2)}$.

- Notes
- Time dummies are included in all equations.
 - Asymptotic standard errors robust to general cross-section and time series heteroskedasticity are reported in parentheses.
 - The GMM estimates reported are all two step except column (a1).
 - The m_2 , Sargan, difference-Sargan and Hausman statistics are all two step versions of these tests except in column (a1). In column (a1) the m_2 and Hausman statistics are asymptotically robust to general heteroskedasticity, whilst the Sargan and difference-Sargan tests are only valid in the case of i.i.d. errors. All Hausman statistics test only the coefficient on m_{it-1} . Degrees of freedom for χ^2 statistics are reported in parentheses.
 - The Wald statistic is a test of the joint significance of the independent variables asymptotically distributed as χ^2_k under the null of no relationship, where k is the number of coefficients estimated (excluding time dummies).
 - The basic instrument set used in columns (a1), (a2) and (b) is of the form

$$Z_t = \begin{bmatrix} n_{1t} & n_{12} & 0 & 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & \Delta x'_{1s} \\ 0 & 0 & n_{11} & n_{12} & n_{13} & 0 & 0 & 0 & \cdots & \Delta x'_{1s} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & n_{1t} & \cdots & n_{17} & \cdots & \Delta x'_{1s} \end{bmatrix} \begin{matrix} 1979 \\ 1980 \\ \vdots \\ 1984 \end{matrix}$$

where x_{it} is the vector of exogenous variables included in the equation. For example, the equation for 1979 in first differences can be written as

$$\Delta m_{1t} = \alpha_1 \Delta m_{1t-1} + \alpha_2 \Delta m_{1t-2} + \Delta x'_{1t} \beta + \Delta v_{1t}$$

For companies on which less than 9 observations are available, the rows of Z_t corresponding to the missing equations are deleted and the missing values of n in the remaining rows are replaced by zeroes.

In columns (c) and (d) Z_t is modified to take the form

$$Z_t = [\text{diag}(n_{1t} \cdots n_{1t-1} \cdots n_{1t-1} \cdots n_{1t-1})' (\Delta x'_{1t} \cdots \Delta x'_{1t})'] \quad (s = 2, \dots, 7)$$

where x_{it} is now the vector of explanatory variables excluding wages and capital but including stacked sales and stocks.

TABLE 3
Employment equations
Alternative estimates

Dependent variable: $\ln(\text{Employment})_{it}$

Independent variables	(e) AHD	(f) AHL	(g) OLS	(h) Within-groups
$n_{it(-1)}$	1.423 (1.001)	2.308 (1.055)	1.045 (0.051)	0.734 (0.058)
$n_{it(-2)}$	-0.165 (0.128)	-0.224 (0.117)	-0.077 (0.048)	-0.141 (0.077)
w_{it}	-0.752 (0.230)	-0.810 (0.283)	-0.524 (0.172)	-0.557 (0.155)
$w_{it(-1)}$	0.963 (0.768)	1.422 (0.851)	0.477 (0.169)	0.326 (0.143)
k_{it}	0.322 (0.105)	0.253 (0.110)	0.343 (0.048)	0.385 (0.056)
$k_{it(-1)}$	-0.325 (0.386)	-0.552 (0.357)	-0.202 (0.064)	-0.084 (0.053)
$k_{it(-2)}$	-0.095 (0.123)	-0.213 (0.145)	-0.116 (0.035)	-0.025 (0.042)
ys_{it}	0.766 (0.311)	0.991 (0.338)	0.433 (0.176)	0.521 (0.193)
$ys_{it(-1)}$	-1.362 (0.881)	-1.938 (0.992)	-0.768 (0.248)	-0.659 (0.208)
$ys_{it(-2)}$	0.321 (0.416)	0.487 (0.425)	0.312 (0.130)	0.001 (0.139)
m_2	-0.781	-0.919	-1.029	
Wald test	199.3 (10)	101.1 (10)		
R^2			0.994	0.689
Number of observations	471	611	751	751

Notes.
(i) Time dummies are included in all equations.
(ii) Asymptotic standard errors robust to general cross-section and time series heteroskedasticity are reported in parentheses.
(iii) The m_2 and Wald tests are asymptotically robust to general heteroskedasticity.
(iv) Columns (e) and (f) report Anderson-Hsiao-type estimates of the equation in first differences: $\Delta n_{it(-1)}$ is treated as endogenous and the additional instruments used are $\Delta n_{it(-3)}$ in (e), and $n_{it(-3)}$ in (f), so that one further cross-section is lost in (e) and the effective sample period becomes 1980-84.
(v) Column (g) reports OLS estimates of the equation in levels, where the effective sample period becomes 1978-1984.
(vi) Column (h) reports within-groups estimates. These are OLS estimates of the equation in deviations from time means.

he short-run effect of product demand fluctuations on labour demand as reflecting the practice of normal cost pricing.

Finally in column (d) we report estimates of the Euler equation model given in (18). Here again we treat wages and capital as endogenous variables. Although the tests for serial correlation remain below their critical values, the coefficient estimates are not favourable to the Euler equation interpretation. The coefficients on capital and industry output are poorly determined, whilst those on the lagged dependent variable imply a real

TABLE A1

Number of records on each company	Number of companies
7	103
8	23
9	14

tests examine overidentifying restrictions. Monte Carlo evidence tells the researcher which tests are reliable in small samples.

7 Short summary

- Difference-GMM removes fixed effects and instruments differenced endogenous variables with lagged levels.
- Validity depends on no serial correlation in level errors.
- The AR(2)/ m_2 test is the key diagnostic.
- GMM has small bias and substantial efficiency gains over Anderson-Hsiao.
- Non-robust tests can be misleading under heteroskedasticity.
- The U.K. employment application illustrates the method and confirms the efficiency gains.

8 Conclusion

8.1 Core conclusion

Dynamic panel models with fixed effects can be estimated using internal lag instruments, but the identifying assumptions must be tested.

8.2 Practical conclusion

Researchers should report AR(2), Sargan/Hansen-type, and difference-Sargan diagnostics and treat two-step standard errors cautiously in small samples.

8.3 Takeaway

Arellano and Bond provide the foundation for modern dynamic panel GMM: internal instruments, efficient weighting, unbalanced-panel implementation, and specification tests.